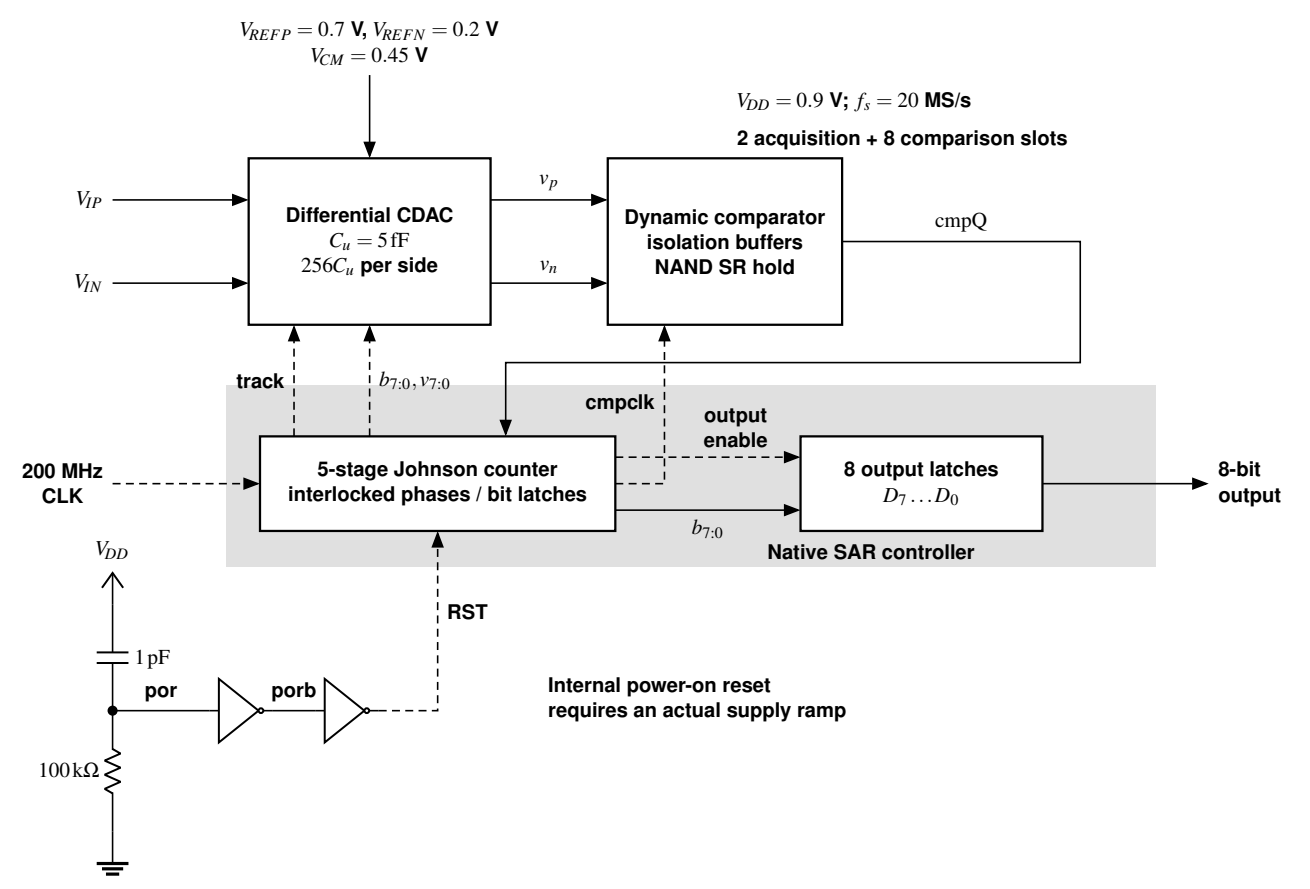

$V_{REFP} = 0.7$ V, $V_{REFN} = 0.2$ V
$V_{CM} = 0.45$ V
$V_{DD} = 0.9$ V; $f_s = 20$ MS/s
2 acquisition + 8 comparison slots
V_{IP}
V_{IN}
Differential CDAC
$C_u = 5$ fF
$256C_u$ per side
v_p
v_n
Dynamic comparator
isolation buffers
NAND SR hold
cmpQ
track
$b_{7:0}, v_{7:0}$
cmpclk
output
enable
200 MHz
CLK
5-stage Johnson counter
interlocked phases / bit latches
8 output latches
$D_7 \ldots D_0$
8-bit
output
$b_{7:0}$
Native SAR controller
V_{DD}
RST
1 pF
por
porb
Internal power-on reset
requires an actual supply ramp
100 kΩ